

# What Matters Becomes Measurable: A Synthesis of Structure-Compatible Generalization, Concern Geometry, and Proxy-Resistant Finite Agents

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Research Derived Experiments

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## Abstract

This synthesis condenses the July 2026 “Metaphysics of Intelligence” archive and the associated Research Derived Experiments repository into one reviewer-facing argument. The central result is not that the work proves consciousness, general agency, or universal OOD robustness. It does something more actionable: it shows how to test whether finite learning systems succeed because the right structure controls behavior, rather than because a proxy passed the surface metric.

Across the archive, the strongest evidence falls into five clusters. First, structure-compatible model selection shows that, under underspecification, compatibility with deployment-relevant transformations can select better OOD models than train or in-distribution validation, with the strongest semantic selection result reaching selected OOD accuracy 0.978 versus 0.919 for ID, train, and random selectors. Second, concern geometry shows that value and viability deform representation: systems allocate resolution, objecthood, and memory toward what future action requires. Third, the minimal-agent line shows which mechanisms make concern load-bearing: action-conditioned consequence models, vector-valued viability heads, identifying interventions, current-error calibration, value-of-information probes, and decision-layer habituation. Fourth, Suite C and the long-horizon moved-bottleneck/tool suite turn agency claims into proxy-resistant benchmarks: behavior alone is not a pass; a pass requires behavior, a structure-specific gate, and anti-cheat controls. Fifth, foundation-model contact is bounded but real: open-model and frozen-encoder results validate selected transport signals, while strict activation-steering gates reject broader semantic-control claims.

The practical takeaway is a design and evaluation rule: for every proposed agent capability, find the proxy that can pass without the intended structure, then add the smallest gate, architecture, or intervention that makes the intended structure load-bearing. This paper prioritizes the evidence, states the strongest safe claims, clusters the experiments, gives operational laws with proof sketches, maps field contributions, identifies the most actionable and commercially valuable directions, and ranks future work.

## 1 What Matters First

Do not read this archive linearly. It contains a research program, a lab notebook, multiple draft papers, negative controls, reviewer-risk audits, and publication packages. The fastest accurate read is:

1. **Core technical contribution:** structure-compatible model selection under underspecification. When train and ID validation cannot tell shortcut and transportable solutions apart, select the model whose learned function preserves the transformations deployment will reuse.
2. **Core benchmark contribution:** no finite-agent suite should pass on behavior alone. Require behavior plus a structure gate plus anti-cheat controls.
3. **Core architecture contribution:** concern is not a scalar reward label. In finite agents it appears as metric allocation, action-conditioned consequence prediction, vector viability, identifying intervention, re-engagement after world change, and commitment at tool/action surfaces.

4. **Core product contribution:** the work is most immediately valuable as an eval and observability layer for model selection, agent reliability, tool commitment, stale-confidence detection, and OOD governance.
5. **Core claim boundary:** the work does not prove consciousness, broad autonomy, production reliability, biological validity, or universal OOD certification. Its strongest claims are finite, controlled, source-backed, and useful.

**The one-sentence thesis.** Finite intelligence becomes measurable when we ask which structure is preserved, which distinctions the system spends capacity on, and whether that structure still controls behavior under shift, intervention, memory pressure, inquiry pressure, and proxy temptation.

**The one-sentence evaluation rule.** A system does not pass because it gets the answer right; it passes when the right variable is represented, attributed, queried, preserved, or committed at the causal surface where action is selected.

## 2 Source Coverage and Claim Discipline

The synthesis is based on the local Metaphysics of Intelligence archive and the repository sources in this worktree. The archive includes the numbered paper sequence from the geometric-meaning seed paper through the July 6 structure, benchmark, Suite C, and open-model packages; the ICML publication package; the Structure-Compatible Generalization folder; Suite C re-engagement and neural transfer folders; Suite C teacher-free inquiry results; Phase Arc 2 and 3 papers; virtual-governor stress-signal notes; and critical reviews.

Where duplicate PDFs and Markdown/TeX sources exist, this paper treats the source-backed repository file and the archived PDF as one evidence item. Where later July 6 versions supersede earlier drafts, the later version supplies the claim boundary and the earlier versions supply lineage. Negative reports and reviewer-risk audits are first-class evidence.

**Claim levels.** The archive contains several kinds of evidence:

- **Strong finite empirical result:** preregistered or gated controlled experiments with negative controls and result rows.
- **Bounded real-model validation:** open-LM, frozen-encoder, or API behavior results that preserve a finite diagnostic.
- **Diagnostic negative:** failed gates that identify a mechanism boundary or reject an attractive interpretation.
- **Synthesis or position result:** architecture laws and field framing derived from the empirical stack.
- **Conceptual seed:** philosophical vocabulary that motivated the tests but is not by itself empirical evidence.

The strongest paper should lead with the first two categories, not with the conceptual seed.

## 3 Cluster Map

| Cluster                               | Question                                                                    | Best evidence                                                                                                                                                          | Current claim                                                         |
|---------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Structure-compatible generalization   | Which model will survive deployment shift when ID evidence ties?            | Weakness/OOD papers; SCG phases 1–6; semantic retrieval and semantic selection; tie-break stress and bootstrap reports.                                                | Strong finite model-selection claim.                                  |
| Concern as metric allocation          | Does value/viability bend representational geometry?                        | Objects from concern; valence tapestry; metric deformation; effective-dimension law; semantic metric deformation.                                                      | Strong controlled claim, bounded real-encoder validation.             |
| Concern to action                     | When does representation become competence?                                 | Delta-E auxiliary learning; two bottlenecks; planning from concern; exploration diagnostics; state-dependent and allostatic results.                                   | Strong mechanism ladder with important negative controls.             |
| Self/world attribution and inquiry    | How does an agent know what came from itself, the world, or stale evidence? | First-order self negative; null intervention; current replay; vector self; world responds; Suite C hand, learned, teacher-free, source-estimate/tool-transfer results. | Strong finite agency diagnostic; open-agent replication still needed. |
| Memory and commitment surfaces        | Does a future-critical variable reach the place where action is committed?  | Future Control Moves Memory; Suite D/E tool commitment; prompt hidden localization; fixed-action localization; causal patching; API behavior.                          | Strong synthetic and bounded open-model/API diagnostic.               |
| Proxy-resistant benchmark methodology | How should agent evals avoid being fooled by final answers?                 | Causally grounded agents benchmark; release schema; benchmark cards; proxy-resistant package.                                                                          | Highly actionable benchmark methodology.                              |
| Foundation-model boundary             | Can hidden semantic geometry steer generation or action?                    | Phase 5/6 transport; semantic-specificity tracker; strict generation/readout gates; Pythia-160M negative replication.                                                  | Bounded transport positives; broad semantic steering not established. |

## 4 Evidence Cluster 1: Structure-Compatible Generalization

### 4.1 What was learned

The OOD/generalization line began with an operational version of weakness: measure how many deployment-relevant transformations a candidate function preserves. The early symbolic result is sharp: in finite cyclic and dihedral settings where shortcuts and invariant rules both fit training data, weakness selects the invariant rule in 100% of trials while the tested simplicity, description length, MDL-style, compression, validation, loss, and flatness proxies select the shortcut. Neural sweeps then show learned-function weakness as a strong OOD correlate.

The newer Structure-Compatible Generalization package broadens this into a model-selection program. The key shift is from “weakness as a diagnostic” to “compatibility as a selector.” Across symbolic, modular, vision, learned generator, finite text, semantic retrieval, and semantic model-zoo selection settings, the useful question is not “which model validates best?” but “which model preserves the transformation family deployment will require?”

### 4.2 Most important results

- Phase 1: true compatibility is the top OOD predictor in modular neural, symbolic cyclic, and vision rotation tasks. In selection without OOD labels, true compatibility selects mean OOD 1.000 across three domains, versus 0.660 for train or ID validation and 0.458 for wrong compatibility.
- Phase 2: discovered compatibility regularization improves high-ID mean OOD from 0.178 to 0.519 and 0.573 at regularization weights 0.05 and 0.20.
- Phase 4 language-template substitution: discovered compatibility correlates with OOD at Pearson  $r = 0.858$ , while wrong compatibility is weak/negative.
- Phase 5 semantic retrieval: true compatibility reaches  $r = 0.940$ , discovered compatibility reaches  $r = 0.861$ , and wrong compatibility is strongly anti-predictive at  $r = -0.872$ .
- Phase 6 semantic model selection: learned compatibility selects OOD 0.978, versus 0.919 for random, train, and ID selectors and 0.751 for wrong compatibility. The learned-minus-random lift is 0.059 with 95% CI [0.052, 0.065]. Mean-tie, worst-tie, and random-tie stress tests all pass.

### 4.3 Why it matters

This is the archive’s cleanest main-conference-style contribution. It addresses underspecification directly: modern pipelines often produce many models with similar train and validation metrics but different deployment behavior. Compatibility gives a practical criterion for choosing among them when the deployment shift has known, inferable, or learnable structure.

### 4.4 Boundary

This is not universal OOD certification. It is a controlled finite result for structured-domain shifts where the relevant transformation family can be specified, weakly inferred, or learned well enough to audit. The vision/random augmentation caveat and phases 1–5 row-regeneration caveats should remain visible.

## 5 Evidence Cluster 2: Concern as Metric Allocation

### 5.1 What was learned

The conceptual seed says meaning is geometry under concern. The empirical version is narrower: finite systems spend representational capacity on distinctions that matter for viability, prediction, or future action.

The object-formation and valence papers show that encoders trained under viability pressure cluster by causal role rather than sensory similarity. Vector-valued valence then shows why scalar reward is not enough: preserving energy and damage as separate predicted consequences supports zero-shot reweighting across hungry, injured, and balanced contexts, while scalar drive collapses distinctions that later matter.

The metric-deformation papers extend the same idea into spatial and semantic representations. Value-weighted training moves local metric density toward high-value regions. The effective-dimension paper is especially useful because it is an honest falsification/repair: the expected physical 2-D allocation law does not hold; the measured exponent implies an effective allocation dimension near one in the tested RNN harness.

### 5.2 Most important results

- Valence encoders group objects by causal reward role, not mere sensory color or label.
- Vector Delta-V heads preserve multi-dimensional concern and support zero-shot priority reweighting where scalar drive heads fail.
- Value-weighted metric deformation moves resolution toward priority fields in controlled models.
- The rate-distortion/effective-dimension test rejects a too-simple 2-D allocation law and reveals the architecture’s effective bottleneck.
- Phase 4 and Phase 6 semantic metric tests show bounded transfer into controlled semantic embeddings and actual frozen encoders, with deformed margin lift 0.409 versus random-label lift 0.008 in Phase 6.

### 5.3 Why it matters

This cluster supplies the scientific bridge between representation learning, interpretability, and cognitive science. It makes “what matters” measurable as resolution allocation, object formation, value-margin movement, and priority-sensitive reweighting. For product development, this is the basis of priority-aware embeddings and retrieval systems: measure whether the embedding space allocates resolution to the distinctions the task will actually reuse.

## 6 Evidence Cluster 3: From Concern to Action

### 6.1 What was learned

Representation is not competence. Several papers exist because the program kept finding separations:

- a representation can encode reward structure while a policy-gradient learner fails to exploit it;
- a policy can succeed while using a proxy representation;
- a model can predict total outcome while misattributing self and world components;

- an uncertainty signal can be high while not indicating useful information gain;
- a quiet probe policy can be stable or falsely calm.

The clean positive mechanism is **predictive policy closure**: an action-conditioned consequence model can become the policy by argmax. In Planning from Concern, a Delta-E model reaches full return and near-perfect action accuracy on the XOR task without optimal-action labels or policy-gradient training.

The negative results are just as important. State-dependent concern fails under smooth online training at the singular boundary. Off-policy state coverage helps but does not solve the boundary. An oracle boundary feature or learned hard gate solves the task. Ensemble uncertainty fails when all models share the same blind spot. Allostatic regulation helps most when it is simple and greedy, not when uncertainty bonuses amplify wrong confidence.

## 6.2 Why it matters

This cluster contributes practical architecture rules. Learn consequences, not only actions. Preserve vector-valued consequences until decision time. Ensure action coverage. Represent hard regime variables explicitly. Treat ensemble variance as a hypothesis, not as epistemic truth.

# 7 Evidence Cluster 4: Self/World Attribution and Adaptive Inquiry

## 7.1 What was learned

The self/world line is one of the archive’s most important scientific arcs. The first result is negative: architectural factorization alone does not recover a self/world decomposition. A self head that sees actions and a world head that does not can still split total prediction arbitrarily while preserving behavior. This is the gauge-symmetry problem.

Null interventions break that gauge. A null action that is dynamically like skip but labeled as a world-anchor reduces false credit dramatically. But the next problem is deciding when to spend costly null probes. Early V-probe targets fail by saturating, tracking stale residuals, or confusing shock noise with systematic error. Current-replay calibration fixes part of the problem by recomputing residuals against the current model. Scale normalization fixes a vector-dimensional calibration failure.

When the world responds to the agent, the decomposition must become richer: direct self, mediated world, and exogenous world. Current error stops being a good proxy for probe value. The agent needs value-of-information. This leads to the Suite C line: re-engage after world change, recover attribution, avoid false calm, spend probes selectively, and reopen after a second shift.

## 7.2 Most important results

- First-order self: factorized heads fail; behavior can be correct while attribution is false.
- Null intervention: active world anchoring reduces false credit; passive exposure is insufficient.
- Current-error calibration: current replay reduces food self MAE close to oracle and beats matched random probing.
- World Responds: three-head direct/mediated/exogenous decomposition is the right structural choice when the world is action-correlated.
- Suite C hand policy: final affected MAE 0.112, first-shift selectivity 16.927, second-shift reopenability 16.667, 22.6 probes, all C1–C6 gates pass.

- Suite C learned transfer: final affected MAE 0.112, first-shift selectivity 16.667, second-shift reopenability 17.448, 23.1 probes; stale, wrong, suppressed, and matched-random controls fail.
- Suite C teacher-free: reward/CEM policy passes C1–C6 plus T1/N1 with final affected MAE 0.095, recovery 1.000, selectivity 12.500, second-shift reopenability about 11, and about 22 probes. A 64-seed bootstrap preserves the pass, with selectivity lift over matched random 11.292 [10.976, 11.596].
- Source-estimate/tool-transfer follow-up: replacing privileged source-identity with an observable estimate and routing through a local JSON-like tool adapter passes, while malformed-tool control fails.

### 7.3 Why it matters

This is the archive’s best agency-eval result. It gives an operational test for world-coupled agency: not just acting, not just remembering, but knowing when the model has lost contact with the world and reopening inquiry selectively. For product development, this points toward a **re-engagement governor** between memory, tools, planning, and action.

## 8 Evidence Cluster 5: Memory and Commitment Surfaces

### 8.1 What was learned

Future Control Moves Memory introduces the long-horizon moved-bottleneck diagnostic. Four early slots are matched for salience and frequency. One becomes future-critical, and the critical slot moves across runs. The question is not only whether the agent answers correctly. It is whether hidden-state sensitivity, generated action, tool call, repair behavior, and causal readout move with the future-critical slot.

The result ladder is unusually complete. The diagnostic passes through a synthetic trained-agent ladder, longer horizons, closed-loop tool handoff, repair, structured tool calls, multi-field schemas, stochastic API failures, 8-slot arguments, aliases, text arguments, generated JSON, autoregressive JSON, prompt-level open models, hidden localization, fixed-action localization, causal patching, prompt-family robustness, and black-box API behavior.

### 8.2 Most important results

- Synthetic trained agents reach 1.000 closed-loop accuracy and positive moved-slot specificity across multiple tool/repair/action surfaces.
- Initial prompt-level final-prompt hidden specificity is a controlled strong negative: behavior passes, but the measured hidden site does not.
- Hidden localization resolves the negative: 17 registered hidden sites pass across Qwen 0.5B, Qwen 1.5B, and SmolLM2 1.7B, strongest at generated final states.
- Fixed-action localization removes the generated-answer-token confound: 24 hidden sites pass even when assistant action tokens are held fixed.
- Fixed-prefix causal patching shows value-prefix hidden states can shift next-token value logits, while prompt-final sites remain negative controls.
- Prompt-family causal patch robustness passes across standard, compact, and audit-checklist framings.

- API black-box behavior: Gemini, Anthropic, and OpenAI pass the matched prompt-family suite. External stress is mixed: Gemini and Anthropic pass, OpenAI GPT-4.1 Nano shows a sparse dispatch failure localized to one reproduced cell out of 16 in the robustness follow-up.

### 8.3 Why it matters

This cluster gives an interpretable and product-relevant memory test. A business agent should not only hold context; it should preserve the variable that will later control a tool call, schema field, repair branch, or emitted value. The long-horizon suite is directly actionable for tool-use reliability, workflow agents, API-call auditing, and context-window memory diagnostics.

## 9 Evidence Cluster 6: Foundation-Model Contact and Rejected Steering

### 9.1 What was learned

The archive contains a tempting story about hidden semantic geometry in language models. The careful version is: hidden geometry often predicts labels or moves log-prob surfaces, but strict behavior, generation, alias, and control gates reject broad semantic steering claims.

Phase 5 is a transport suite. It shows that controlled proxy mechanisms survive harder model-like gates: language-action transport passes in a stronger proxy, semantic metric deformation survives a foundation-style encoder proxy, role-routed heads break the mediated-identifiability ceiling, and seam consistency, not topology alone, carries OOD lift.

Phase 6 moves to actual open LMs and frozen encoders. It reports weak positive open-LM action-coupling signals across three decoder LMs, with Qwen 0.5B strongest and Pythia 70M weak/control-like. Frozen-encoder metric deformation passes with deformed margin lift 0.409 versus random 0.008.

The activation-steering line is different. The semantic-specificity tracker records many failed rescues: held-out aliases, target-disjoint controls, random relation nulls, generation-match gates, learned generation readout, yes-bias controls, residualization, whitening, pair-specific optimization, feature masks, state gates, multi-class prototype gates, and Pythia-160M replication all prevent a broad steering claim. The current safe statement is diagnostic: the strict verifier exposes a narrow specificity frontier, but the frontier has not shown stable held-out alias specificity on expanded controls.

### 9.2 Why it matters

This is exactly how a healthy research program should behave. It promotes bounded transport signals while rejecting overbroad mechanistic claims. For future research, exact-relation readout identifiability should come before more steering optimization on the same brittle interface.

## 10 Operational Laws and Proof Sketches

**Operational law 1** (Behavior is not load-bearing structure). *Final accuracy, return, or tool success is not sufficient evidence that a system used the intended structure.*

**Evidence sketch.** The archive repeatedly constructs behavior-only passes that fail structure gates: shortcut OOD rules, gauge-symmetric self/world heads, visible-control memory cases, signal-suppression false calm, and activation directions that move positives and controls together. The proxy-resistant benchmark rule is the repair: behavior plus structure gate plus anti-cheat control.



**Empirical proposition 1** (Structure-compatible selection). *In finite structured shifts, compatibility with the deployment-relevant transformation family can improve OOD-free model selection among train/ID-tied candidates.*

**Evidence sketch.** The evidence ladder runs from oracle finite transformations through inferred and learned generators, finite text substitutions, frozen semantic retrieval, and semantic model-zoo selection. Phase 6 selects OOD 0.978 with discovered compatibility versus 0.919 for train/ID/random and 0.751 for wrong compatibility, with bootstrap and tie-break stress passing. The boundary is finite structured shifts, not universal OOD certification.

**Operational law 2** (Predictive policy closure). *When an agent learns an action-conditioned consequence model for what it cares about, the consequence model can become the policy by action argmax.*

**Evidence sketch.** Planning from Concern shows that a Delta-E consequence model solves the XOR homeostatic task with near-perfect action accuracy and full return, without optimal-action labels or policy-gradient training. Earlier Delta-E auxiliary and two-bottleneck papers show why this matters: representation alone is not competence, and policy-gradient learning can fail to exploit a useful representation.

**Operational law 3** (Vector concern must survive until decision time). *Scalar reward collapses distinctions that later priorities may need.*

**Evidence sketch.** The valence tapestry and vector first-order self papers show that vector Delta-V heads support zero-shot reweighting across hungry, injured, and balanced contexts, especially for medicine handling. Scalar drive heads cannot preserve the needed distinctions.

**Operational law 4** (Attribution needs identifying interventions). *Architectural factorization is not self/world identification.*

**Evidence sketch.** First-Order Self shows that self/world heads can split total prediction arbitrarily while preserving behavior. Null interventions, source labels, temporal asymmetry, or interventional contrast are needed to break the gauge. Null anchoring and later contrastive/role-routed mechanisms supply positive evidence.

**Operational law 5** (Probe value is not current error). *The best next question is not necessarily where the current model is most wrong.*

**Evidence sketch.** World Responds shows an oracle-current-error probe can be much worse than a learned probe because high error may be irreducible or poorly positioned for future model improvement. Phase 4's learned VOI gate passes while current error and ensemble variance fail. Suite C strengthens this into re-engagement under world change.

**Operational law 6** (Cool the decision, not the signal). *Habituation should reduce unnecessary action while preserving surprise.*

**Evidence sketch.** Habituated Re-engagement shows decision-layer cooling reduces anxious probing, while signal-layer cooling creates false calm. Suite C formalizes the no-false calm gate: probe rate, uncertainty signal, and outcome error must improve together.

**Operational law 7** (Memory matters at commitment surfaces). *Memory becomes agent-relevant when a future-critical variable reaches an action, tool, schema, repair, emitted value, or causal readout surface.*

**Evidence sketch.** The moved-bottleneck suite shows memory sensitivity follows the moved critical slot through hidden states, generated JSON, tool calls, repair branches, and causal value-token patches. Visible-control and prompt-final negative controls rule out several shortcut explanations.

## 11 Field Contributions

### 11.1 Machine learning generalization

The contribution is a model-selection criterion for underspecification. If many models tie on train and ID validation, select by compatibility with the deployment-generating structure. This directly targets shortcut learning, domain generalization, and OOD model selection.

### 11.2 Agent evaluation and AI safety

The benchmark contribution is the minimum pass rule. A capable-looking agent should not pass unless the right causal/structural reason explains its behavior. This connects to reward hacking, goal misgeneralization, stale confidence, tool-call brittleness, and proxy optimization.

### 11.3 Interpretability

The work shifts interpretability from “is information present?” to “does the information control behavior at the right surface?” The memory, attribution, causal patch, and activation-specificity lines are all examples of this surface-based standard.

### 11.4 Agent architecture

The archive suggests concrete modules: action-conditioned consequence models, vector viability heads, source-attribution gates, null or contrastive interventions, value-of-information probe policies, re-engagement governors, role-routed heads, and commitment-surface tests.

### 11.5 Cognitive science, neuroscience, and philosophy

The contribution is not a proof of consciousness. It is a computational operationalization of concern-like organization: what matters bends representation, inquiry, action, attribution, and memory. This gives philosophy and cognitive science a testable vocabulary while keeping biological and human claims out of scope.

## 12 Actionability, Ironcladness, and Commercial Value

| Area                                 | How ironclad?                                                                     | How actionable?                                              | Why lucrative?                                                                                        |
|--------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Structure-compatible model selection | High within finite structured shifts; Phase 6 has bootstrap and tie-break stress. | Add compatibility selectors to model-zoo and eval pipelines. | Better model selection under deployment risk is immediately valuable for regulated and enterprise ML. |

| Area                             | How ironclad?                                                                        | How actionable?                                                                             | Why lucrative?                                                                              |
|----------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Proxy-resistant agent benchmarks | High as methodology; suites vary by maturity.                                        | Ship benchmark cards, JSONL rows, and one-command smoke tests.                              | Every agent platform needs evals that catch shortcut success and stale confidence.          |
| Suite C re-engagement governor   | High in finite harness; open-agent transfer pending.                                 | Add stale-belief monitors and selective inquiry policies between memory and action.         | Reduces costly agent mistakes from stale assumptions, especially in long-running workflows. |
| Commitment-surface memory tests  | High across synthetic, open-model, and API behavior surfaces, with clear non-claims. | Test whether critical variables survive tool calls, schemas, repairs, and generated values. | Tool-use and workflow reliability are direct product pain points.                           |
| Semantic metric deformation      | Medium-high; controlled and frozen-encoder evidence is positive.                     | Build priority-aware retrieval or embedding audits.                                         | Retrieval quality and high-value-case recall are commercially important.                    |
| Activation steering              | Low for broad claims; high as a diagnostic negative.                                 | Stop optimizing brittle global vectors; test exact readout first.                           | Avoids shipping unsafe or fake controls; useful for research tooling, not product yet.      |
| Concern-geometry synthesis       | Medium as theory; depends on narrower papers.                                        | Use as architecture vocabulary after empirical anchors.                                     | Valuable for strategy, but not a standalone product proof.                                  |

## 13 Product Development Implications

**1. Model selection product.** Build an OOD-free selector that evaluates compatibility against known or inferred deployment transformations. Inputs: candidate models, deployment transform hypotheses, train/ID filters, wrong-transform controls. Output: selected model, regret estimate, compatibility audit, claim boundary.

**2. Agent reliability product.** Build a proxy-resistant eval suite with vector scores: behavior, structure, attribution, inquiry, commitment, and generalization. The minimum pass rule should be a product primitive, not a paper footnote.

**3. Re-engagement governor.** Add a layer between memory, tools, and action that tracks assumption staleness, world-change signals, affected-source estimates, cost-aware inquiry, and no-false-calm recovery. The first product version can be benchmark-only; the second can become runtime monitoring.

**4. Tool commitment observability.** Instrument agents to verify that future-critical variables survive prompt state, tool-call JSON, schema validation, repair branches, and emitted values. The long-horizon suite is a direct template.

5. **Priority-aware retrieval.** Use metric-deformation audits to determine whether retrieval/embedding systems allocate resolution to high-value business distinctions, not only average semantic similarity.

## 14 Prioritized Future Directions

1. **External Suite C replication.** Highest scientific and product priority. Reproduce teacher-free/source-estimate/tool-transfer Suite C in an open LM or API-agent harness. This would move the inquiry governor from a finite simulator result to a broadly legible agent-eval result.
2. **Publish Structure-Compatible Model Selection.** Keep Phase 6 as the headline, include row-level artifacts, bootstrap intervals, tie-break stress, formal compatibility definitions, and a strict finite-structured claim boundary.
3. **Release proxy-resistant benchmark package.** Package Suite C and Suite D/E with public JSONL rows, benchmark cards, fixture smoke tests, release schema, and a Hugging Face dataset-style card.
4. **Productize commitment-surface memory tests.** Turn the moved-bottleneck/tool suite into a reusable CI/eval for tool-using agents.
5. **Build role-routed self/world architectures.** Phase 4 suggests the mediated-identifiability ceiling is architectural. Test disjoint per-role heads, mixture-of-experts, and richer counterfactual interventions in larger agent environments.
6. **Turn compatibility into training, not only selection.** Extend discovered compatibility regularization to practical domains where deployment transforms are partially known.
7. **Advance priority-aware embedding audits.** Move semantic metric deformation from post-hoc/frozen evaluation toward controlled finetuning or adapter training with business-relevant priority fields.
8. **Do exact-relation readout before more steering.** The activation line should first establish stable exact-relation identifiability under strict controls before attempting stronger generation steering.
9. **Keep the broad concern-geometry paper as synthesis.** It should follow the narrower empirical submissions. It is valuable, but leading with it risks reviewer confusion.

## 15 What Not To Prioritize

- Do not lead with consciousness, AGI, or metaphysics in technical venues.
- Do not claim universal OOD certification.
- Do not claim production agent reliability from finite Suite C or black-box API behavior alone.
- Do not promote activation steering as a positive result yet.
- Do not collapse benchmark results into a single leaderboard score.
- Do not hide negative controls; they are the reason the program is credible.

## 16 Recommended Paper and Release Sequence

1. **Structure-Compatible Model Selection Under Underspecification.** Main-conference target. Most complete, most legible, strongest statistical anchor.
2. **Proxy-Resistant Benchmarks for Causally Grounded Finite Agents.** Benchmark target. Lead with the minimum pass rule and Suite C plus Suite D/E.
3. **Teacher-Free Re-Engagement.** Focused agency paper. Stronger if external open-agent replication is added.
4. **What Matters Becomes Measurable.** Broad synthesis/position paper. Use after the narrower empirical anchors exist publicly.

## 17 Conclusion

The archive’s deepest contribution is not any single experiment. It is a research discipline: whenever a system appears capable, ask what proxy could make the behavior pass without the intended structure. Then make the structure load-bearing, add an anti-cheat control, and bound the claim to the weakest validated level.

That discipline produced several strong results. Structure-compatible model selection gives a practical answer to underspecification. Suite C gives a finite test of adaptive inquiry after world change. The moved-bottleneck suite shows how future-control relevance moves memory and tool commitment. The metric-stack line shows how concern deforms representation and where architecture ceilings appear. The activation-steering failures show that the program can reject an attractive story when the controls do not support it.

If one sentence should survive into future research and product work, it is this: **Do not ask only whether the agent succeeded; ask whether the future-controlling structure reached the surface where success was chosen.**

## A Primary Reading Spine

The shortest serious reading order is:

1. ICML publication package README.
2. Structure-compatible model-selection result tables.
3. Semantic-selection control paper.
4. Causally grounded finite-agent benchmark paper.
5. Suite C teacher-free inquiry package README.
6. Suite C teacher-free wide-seed bootstrap report.
7. Long-horizon moved-bottleneck paper and benchmark card.
8. Metric Stack synthesis paper.
9. Paper-readiness and semantic-specificity audit notes.

## B Source Clusters Used

- **Conceptual and metric-stack archive:** [0\\_geometric\\_meaning\\_and\\_agency.pdf](#); [26\\_Metric\\_Stack\\_of\\_Concern\\_v4\\_2026\\_07\\_06.pdf](#); [papers/metric\\_stack\\_synthesis/paper.md](#); metric-stack position package.
- **Weakness and structure-compatible generalization:** [1\\_Weakness\\_Predicts\\_OOD\\_2026\\_06\\_09.pdf](#); [30\\_Weakness\\_Predicts\\_OOD\\_2026\\_07\\_02\\_draft.pdf](#); weakness-invariance paper; structure-compatible generalization papers, results, and ICML package.
- **Concern/action/mechanism ladder:** numbered papers 4–25 in the archive; Planning from Concern; First-Order Self; Null Intervention; World Responds; Probe-Value Re-engagement; and Habituated Re-engagement.
- **Suite C:** Suite C re-engagement archive; neural probe transfer archive; teacher-free inquiry package; and world-responds result reports.
- **Long-horizon and tool commitment:** [31\\_Future\\_Control\\_Moves\\_Memory\\_2026\\_07\\_06.pdf](#); long-horizon moved-bottleneck paper; benchmark card; and result reports.
- **Foundation-model and activation boundary:** Phase 4, Phase 5, and Phase 6 archive papers; paper-readiness and semantic-specificity audits; and activation-geometry result reports.
- **Benchmark packaging and release:** Causally grounded agents benchmark paper; benchmark docs; release schema; publication sharing map; and proxy-resistant agent benchmark package.

## C Safety and Non-Claims

This synthesis intentionally avoids the following claims:

- consciousness, sentience, or human-like subjective experience;
- biological or neural validity beyond computational analogy;
- production reliability for deployed agents;
- universal OOD certification;
- open-world semantic robustness;
- broad activation-steering control.